

Leveraging Gauge Freedom for Learning Non-Gradient Population Dynamics of Stochastic Systems

Jules Berman^{* 1} Tobias Blickhan¹ Benjamin Peherstorfer¹

Abstract

In existing works on population dynamics inference, there is a focus on flows arising from vector fields that are the gradients of scalar potentials. Among all admissible flows that are compatible with the population dynamic, gradient flows are optimal in a specific sense: they minimize kinetic energy. The selection of fields based on different criteria corresponds to a gauge freedom when determining population dynamics, which we leverage in this work. We propose Non-Gradient Inference Flows (NGIF), an algorithm to infer non-gradient population dynamics using a weak formulation of the continuity equation. This allows us to parameterize with general vector fields as well as choose other selection criteria beyond minimal energy. We demonstrate on a variety of low- and high-dimensional physics problems that this more general approach improves distributional accuracy over gradient-restricted baselines and better captures non-potential transport.

1. Introduction

We aim to learn models of dynamical systems observed through independent samples at each time point t : for each t we are given a set of samples drawn from the time-marginal distribution of the states, but we do not observe trajectories or any correspondence of individual samples across times. The resulting task is to infer population-level dynamics, i.e., a dynamical model whose induced evolution matches the observed sequence of time-marginal laws rather than individual trajectories.

This setting is relevant for stochastic, chaotic, or turbulent systems in physics and engineering, where individual trajectories are intrinsically unpredictable. In such regimes, matching time marginals opens the door to learning pre-

dictive data-driven dynamical and reduced models (Neklyudov et al., 2023; Berman et al., 2024; Blickhan et al., 2025). This setting is also examined in biology, for example in single-cell experiments where destructive measurements provide cross-sectional snapshots along developmental time (Bunne et al., 2022; Lavenant et al., 2023).

Setup and population dynamics inference The setup we consider is that we have samples $x_t \in \mathcal{X} \subseteq \mathbb{R}^d$ over time $t \in [0, T]$ with law $\rho_t \in \mathcal{P}(\mathcal{X})$. We assume that the law admits a density function, which we also denote as ρ_t . The goal of population dynamics inference is to learn a time-dependent velocity field $u_t : \mathcal{X} \rightarrow \mathbb{R}^d$ whose induced flow transports ρ_0 to the marginals $(\rho_t)_{t \in [0, T]}$. The requirement of matching the time marginals can be expressed as asking the pair (u_t, ρ_t) to satisfy the continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t u_t) = 0, \quad (1)$$

in the sense of distributions. Given such a field with sufficient regularity, one can generate representative dynamics by integrating the flow ODE

$$\frac{d}{dt} x_t = u_t(x_t), \quad x_0 \sim \rho_0, \quad (2)$$

which produces a trajectory with $x_t \sim \rho_t$ for all times t .

Gauge freedom A central difficulty is that the continuity equation (1) provides only an underdetermined constraint on the velocity field u_t . The continuity equation only depends on u_t through $\nabla \cdot (\rho_t u_t)$. All dynamics induced by the velocity that do not change this quantity have no effect on the evolution of the time marginals and are therefore not fixed by asking u_t to satisfy the continuity equation. Concretely, if u_t satisfies (1) and w_t is such that $\nabla \cdot (\rho_t w_t) = 0$ in the sense of distributions, then $u_t + w_t$ yields the same evolution of the time marginals ρ_t . This underdetermination of the u_t is a gauge freedom: the time marginals determine only parts of the flow, leaving ρ_t -weighted divergence-free components unconstrained. One can see this as a ρ_t -weighted Hodge-Helmholtz decomposition (Chorin & Marsden, 1993, Page 37) that splits the space $L^2(\rho_t; \mathbb{R}^d)$ into two orthogonal subspaces; the continuity equation only prescribes the solution on one of them.

¹Courant Institute of Mathematical Sciences, New York University, New York, NY 10012, USA. Correspondence to: Jules Berman <jmb1174@nyu.edu>.

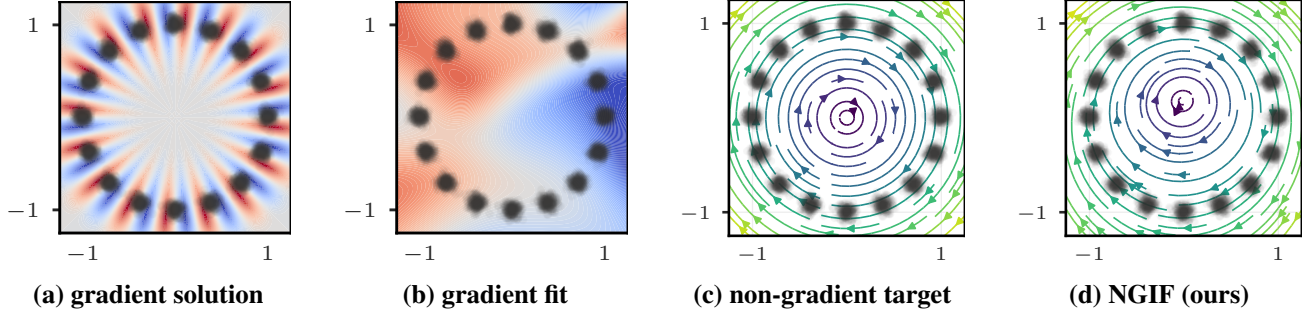


Figure 1. We compare various solutions to the Gigli example A in which a concentric set of Gaussians rotate at a constant speed from left to right. The analytic solution when constrained to using gradient fields in (a) gives a highly irregular target which may be difficult to learn or represent, causing inaccurate approximation of this potential in (b) when we parametrize with a gradient field. In contrast the unconstrained vector field solution in (c) is easily and accurately approximated with our non-gradient flow in (d). Notice additionally, while the solution in (a) is optimal in terms of kinetic energy the solution in (c) has only a marginally larger kinetic energy. Yet for this small improvement, one pays a drastic cost in terms of the regularity of the target.

Fixing the gauge with minimal kinetic energy One canonical choice to fix the gauge is to select the field that has minimal kinetic energy, i.e. setting the solution to zero on the undetermined subspace:

$$\min_{u \text{ satisfies (1)}} \frac{1}{2} \int |u_t(x)|^2 \rho_t(x) dx dt.$$

Selecting the minimal kinetic energy field is closely related to optimal transport (OT) and, under regularity assumptions, yields a unique velocity field that is necessarily a gradient field $u_t = \nabla s_t$ for some potential $s_t : \mathbb{R}^d \rightarrow \mathbb{R}$ (Ambrosio et al., 2005, Proposition 8.4.5).

Much of the existing methodology on learning population dynamics hard-encodes this gauge by parametrizing a potential s_t and learning $u_t = \nabla s_t$ via s_t , as in action matching (Neklyudov et al., 2023; Berman et al., 2024) and the discrete inverse continuity equation method (Blickhan et al., 2025). Methods concerned with unbalanced problems and growth functions (Zhang et al., 2025b;a; Sun et al., 2025) use this parameterization. Approaches that build on the Jordan-Kinderlehrer-Otto framework (Jordan et al., 1998) such as JKONet and related works (Bunne et al., 2022; Alvarez-Melis et al., 2021; Terpin et al., 2024; Pershianov et al., 2025) also restrict the learning to these gradient-type dynamics; see Section 4.

Going beyond gradient dynamics and minimal-kinetic-energy gauges matters While the minimal kinetic energy field provides a principled default, it is only one gauge choice. In particular, it can impose an inductive bias that is at odds with the underlying dynamics when rotational components are essential, such as in Hamiltonian systems or incompressible fluid dynamics. This failure case is also mentioned in (Terpin et al., 2024, Section G.2).

Because the continuity equation (1) constrains only the compressible part of the flow, selecting the minimal kinetic

energy gauge suppresses rotation by construction. In extreme cases, enforcing a gradient structure can turn simple rotational motion into highly oscillatory, pathological potentials whose gradients match the motion only on (or near) the support of ρ_t (Gigli, 2011). Such targets are difficult to represent and learn (Figure 1), and small approximation errors in a highly oscillatory potential can produce large deviations in the generated dynamics.

In addition, hard-encoding $u_t = \nabla s_t$ by parametrizing the potential s_t often complicates optimization because one must backpropagate through spatial derivatives of the network output and because the parameterization restricts the hypothesis class in ways that are not aligned with the data.

The literature on learning non-gradient dynamics is scarce. The recent work Petrović et al. (2025) targets learning non-gradient flows by introducing a prescribed non-gradient reference flow as a prior in the form of velocity information at the sample level. In contrast, we do not require a reference flow or sample velocities. Instead we expose the gauge degree of freedom directly, allowing to select from a broad family of gauge regularizers, including but not limited to minimal kinetic energy.

A related approach is to endow a flow-matching model with a non-trivial metric structure to obtain curved sample trajectories. This is a slightly different problem as these works typically operate in the two-marginal setting. We refer to (Scarvelis & Solomon, 2023) and (Kapuńiak et al., 2024, Section 6) for a review.

Our approach: non-gradient flow inference (NGIF) with explicit gauge regularization We propose a non-gradient approach to population dynamics inference that matches the time-marginal evolution through a moment-based (weak) form of the continuity equation, without restricting the velocity field to gradient form.

For a set of test functions, we require that the rate of change of their expectations under the time marginals agrees with what the candidate velocity field predicts via transport. This yields an objective that can be empiricalized from only samples of the time marginals and with random Fourier feature test functions that lead to an informative and computationally cheap empirical loss.

The key feature of our approach is that the weak continuity equation loss separates time-marginal matching from velocity field selection. By construction, our loss only constrains the part of the field that affects the time marginals, leaving the gauge freedom corresponding to divergence-free components untouched. Rather than resolving this non-uniqueness implicitly by restricting the parametrization (e.g., parametrizing the potential instead of the velocity field), we use explicit gauge regularizers. Importantly, the regularizer is a free design choice to support informative inductive bias beyond minimal kinetic energy such as encouraging smoothness and controlling divergence or curl. This enables learning non-gradient fields that can better reflect dynamics with substantial rotational structure, avoiding the pathological behavior that can arise when rotation is mimicked by rapidly changing time-dependent gradient fields. To our knowledge, our approach is the only population dynamics inference method which allows one to explicitly leverage this gauge freedom.

Contributions (a) We separate matching time marginals from gauge fixing by formulating population dynamics inference through weak continuity equation constraints and addressing non-uniqueness with an explicit, freely chosen gauge regularizer.

(b) We instantiate the weak constraints with random Fourier feature test functions, yielding a scalable loss.

(c) We enable learning non-gradient velocity fields that improve representability and stability, and compared to gradient-only formulations lead to more accurate population dynamics inference.

2. Inferring non-gradient fields

Motivated by the need to allow non-gradient dynamics, we propose to learn a velocity field u_t via the weak form of the continuity equation, which allows us to leverage the gauge freedom to impose structural bias on the field that we want to learn—other structural bias than gradient dynamics.

2.1. Weak continuity equation as a moment equation

We seek a velocity field u_t such that the time marginals ρ_t satisfy the continuity equation (1) in the sense of distributions. Let therefore ϕ be a test function taken from a test

space Φ so that the weak form of (1) is

$$\int \phi(x) (\partial_t \rho_t(x) + \nabla \cdot (\rho_t(x) u_t(x))) dx = 0.$$

for all $t \in [0, T]$. Assuming standard regularity to justify differentiation under the integral, we pull out the time derivative in the first part and integrate by parts in the second term to obtain $\int \phi \nabla \cdot (\rho_t u_t) = - \int \nabla \phi \cdot u_t \rho_t$ and

$$\frac{d}{dt} \int \phi(x) \rho_t(x) dx = \int \nabla \phi(x) \cdot u_t(x) \rho_t(x) dx. \quad (3)$$

If $\mathcal{X} = \mathbb{T}^d$, then there is no boundary and the boundary term vanishes identically for all smooth periodic $\phi \in C^\infty(\mathbb{T}^d)$. If $\mathcal{X} = \mathbb{R}^d$, then the boundary term vanishes under integrability and tail assumptions, e.g., that $\rho_t u_t \in L^1(\mathbb{R}^d)$ with test functions $\phi \in C_b^1(\mathbb{R}^d)$. Recall that $C_b^1(\mathbb{R}^d) = \{\phi \in C^1(\mathbb{R}^d) \mid \|\phi\|_\infty < \infty, \|\nabla \phi\|_\infty < \infty\}$ is the space of all continuously differentiable functions with domain $\mathbb{R}^d$ with bounded value and bounded first derivative. The condition $\rho_t u_t \in L^1(\mathbb{R}^d)$ implies finite momentum in the system, which is often a natural assumption.

Under these conditions, we can write (3) as

$$\frac{d}{dt} \mathbb{E}_{x \sim \rho_t} [\phi(x)] = \mathbb{E}_{x \sim \rho_t} [\nabla \phi(x) \cdot u_t(x)], \quad (4)$$

for test functions $\phi \in \Phi$. The quantity $\mathbb{E}_{\rho_t} [\phi(x)]$ is a generalized moment of ρ_t in the sense that it is the integral of ρ_t against a test function ϕ . Ordinary moments are recovered as the special case that the test functions are monomials, e.g., $\phi(x) = x$ gives the mean, and $\phi(x) = xx^\top$ gives the second moment. We allow a richer set of test functions Φ in the following. Furthermore, the moment equation (4) does not just constrain a static expectation at a single time but it specifies how the time derivative of each generalized moment $\frac{d}{dt} \mathbb{E}_{\rho_t} [\phi]$ in terms of ρ_t and velocity u_t evolves.

2.2. Minimizing weak-form residual

We now operationalize the weak moment identity by enforcing it in expectation over test functions with features sampled from a feature distribution ν .

Let Φ denote an admissible test space and let ν be a probability measure so that $\phi_\omega \in \Phi$ with $\omega \sim \nu$ (see forthcoming Section 2.4). We introduce a residual

$$R_t(\phi; v_t) = \frac{d}{dt} \mathbb{E}_{x \sim \rho_t} [\phi] - \mathbb{E}_{x \sim \rho_t} [\nabla \phi \cdot v_t]$$

In the ideal case where (ρ_t, u_t) satisfies the continuity equation in weak form, we have $R_t(\phi; u) = 0$ for all $\phi \in \Phi$ and $t \in [0, T]$.

We can also build on the Fokker-Planck equation in which case one ends up with the additional term $\frac{\varepsilon^2}{2} \mathbb{E}_{\rho_t} [\Delta \phi]$ which

is added to the residual, ε is an optional hyperparameter. See Appendix F.1 for details.

We can turn this into a loss by minimizing the residual norm over the random test functions

$$\mathcal{L}(v) = \int_0^T \mathbb{E}_{\omega \sim \nu} [|R_t(\phi_\omega, v_t)|^2] dt \quad (5)$$

Minimizing the loss $\mathcal{L}(v)$ drives the residual R_t to be orthogonal to the test space. If having a minimizer with $\mathcal{L}(u) = 0$ implies that u is compatible with the continuity equation depends on the richness of the test space that is covered by features sampled from ν . In particular, we require that the sampled test functions are rich enough to determine the weak form on Φ , i.e., they need to generate a dense subset of Φ , using the fact that $\phi \mapsto R_t(\phi; v)$ is continuous. For the torus, when ν has full support on $\mathbb{Z}^d$, this is sufficient. For $\mathcal{X} = \mathbb{R}^d$, we find an analogous situation for ν Gaussian on $\mathbb{R}^d$.

2.3. Leveraging gauge freedom

The weak continuity constraints determine the velocity field only up to components that do not affect the time-marginal evolution. As a result, there is freedom to impose additional structural bias. While other methods (see Section 1) often are restricted to structural bias given by minimal kinetic energy, our approach allows full flexibility.

The continuity equation has gauge freedom The weak form (and hence the loss $\mathcal{L}$) depends on u_t only through the weighted divergence $\nabla \cdot (\rho_t u_t)$. Equivalently, it only constrains the action of u_t through the flux $\rho_t u_t$ appearing in the continuity equation. Consequently, if u_t is compatible with the weak continuity equation (e.g. achieves $\mathcal{L}(u) = 0$), then so is any perturbed field $u_t + w_t$ such that $\nabla \cdot (\rho_t w_t) = 0$ (in a weak sense), i.e., for any test function $\phi \in \Phi$,

$$\mathbb{E}_{\rho_t} [\nabla \phi \cdot (u_t + w_t)] = \mathbb{E}_{\rho_t} [\nabla \phi \cdot u_t],$$

because $\mathbb{E}_{\rho_t} [\nabla \phi \cdot w_t] = -\langle \phi, \nabla \cdot (\rho_t w_t) \rangle = 0$. Therefore $u_t + w_t$ yields the same weak-form residuals and achieves the same loss. This non-uniqueness of the loss is induced by the natural gauge freedom of the continuity equation, because the continuity equation specifies u_t only up to the non-divergence-free part.

Leveraging the gauge Recovering a unique velocity field requires an additional selection principle or gauge fixing. Concretely, we select u as the solution of

$$\min_u \mathcal{L}(u) + \lambda \mathcal{G}(u), \quad (6)$$

where $\mathcal{G}$ varies depending on the selected gauge.

Uniqueness versus non-uniqueness For certain $\mathcal{G}$ (strictly convex and coercive with $\lambda > 0$), the objective of (6) determines a unique u . However, in general, uniqueness is not required. The role of the gauge regularizer is to impose a structural (inductive) bias on admissible solutions. We will show in our numerical experiments that in many practically relevant settings, we intentionally employ gauge regularizer that are non-coercive and therefore do not yield uniqueness. Even when the optimization problem (6) has multiple minimizers, the regularizer restricts the solution set in a principled way by promoting specific structure. For instance, penalizing divergence $\nabla \cdot u$ typically does not lead to an optimization problem (6) with a unique solution, yet it encourages approximately incompressible behavior and leads to empirically desirable dynamics as our numerical experiments show.

Proposed gauge regularizer In the following, we consider three Gauge regularizers. First, we naturally consider a kinetic energy regularizer,

$$\mathcal{G}_{\text{KIN}}(v) = \frac{1}{2} \int \|v_t\|^2 \rho_t(x) dx dt. \quad (7)$$

Importantly, this means using the kinetic energy as a soft gauge-fixing term rather than being enforced by restricting the learning to gradient fields only, which as shown in Section A can lead to challenging optimization problems. Second, we can penalize the antisymmetric part of the Jacobian of the velocity field, which measures rotation and non-gradient components of the flow,

$$\mathcal{G}_{\text{CURL}}(v) = \frac{1}{2} \int \|\nabla v_t(x) - (\nabla v_t(x))^\top\|_F^2 \rho_t(x) dx dt. \quad (8)$$

Using $\mathcal{G}_{\text{CURL}}$ penalizes rotation and biases the solution toward gradient-like dynamics but still avoids an explicit potential parametrization that would hard-code gradient-only dynamics. Third, we consider the div regularizer,

$$\mathcal{G}_{\text{DIV}}(v) = \int |\nabla \cdot v_t|^2 \rho_t(x) dx dt, \quad (9)$$

which promotes approximately incompressible velocity fields. It is especially useful in fluid-like transport (e.g., tracer particles in a fluid flow field).

2.4. Choosing a test space via random Fourier feature test functions

The key ingredient is constructing a finite number of test functions $\{\phi_i\}_{i=1}^M$ for which the weak continuity equation yields informative low-variance constraints and for which the required derivatives are tractable. We consider Fourier modes $x \mapsto \exp(i\omega^\top x)$ for $x, \omega \in \mathbb{R}^d$, with real and imaginary parts $\cos(\omega^\top x)$ and $\sin(\omega^\top x)$.

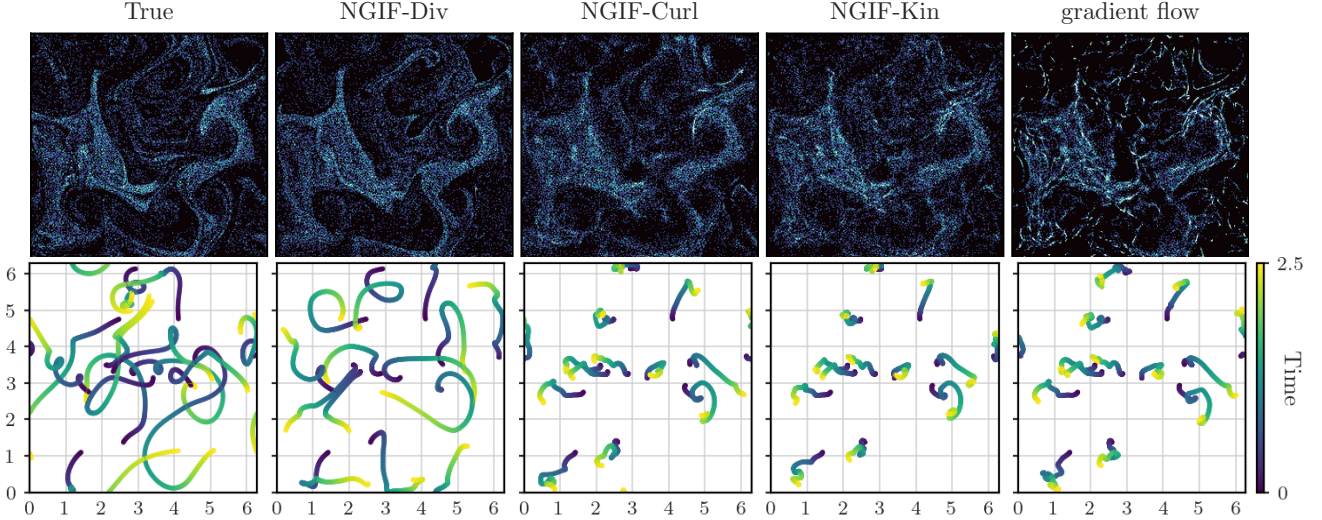


Figure 2. Results for the tracer particle case. We see that the divergence regularizer (9) leads to both the most faithful reconstruction of the particle density (top row) and qualitatively similar sample trajectories (bottom row). Note that the physical sample velocity field is divergence-free in this case.

The appropriate choice of frequencies ω depends on the spatial domain: on a periodic domain $\mathcal{X} = \mathbb{T}^d = [0, 2\pi)^d$ one must use integer frequencies $\omega = k \in \mathbb{Z}^d$ so that $x \mapsto \exp(ik^\top x)$ (and hence $\sin(k^\top x)$, $\cos(k^\top x)$) is well-defined and 2π -periodic in each coordinate. In this case the Fourier modes belong to $C^\infty(\mathbb{T}^d)$ and form the canonical trigonometric dictionary on the torus. On $\mathcal{X} = \mathbb{R}^d$ one may choose ω from $\mathbb{R}^d$. These functions for $\omega \in \mathbb{R}^d$ are C^∞ . Furthermore, for any fixed ω , $\cos(\omega^\top x)$ and $\sin(\omega^\top x)$ belong to $C_b^1(\mathbb{R}^d)$ with $\|\phi\|_\infty \leq 1$ and $\|\nabla \phi\|_\infty \leq \|\omega\|$.

Random Fourier feature test functions We approximate these test spaces using random features. We use paired real features

$$\sin(\omega^\top x), \quad \cos(\omega^\top x), \quad (10)$$

where $\omega \sim \nu$ is sampled from a chosen frequency distribution ν . Pairing $\sin$ and $\cos$ (equivalently, using both real and imaginary parts of $e^{i\omega^\top x}$) yields a lower-variance estimator than single complex features and is standard in random-feature approximations of shift-invariant kernels (Sutherland & Schneider, 2015).

Multiscale random Fourier feature test functions To generate random Fourier feature test functions, we sample frequencies across multiple bandwidths to improve sensitivity to structure across multiple resolutions in the data as in (Parra & Tobar, 2017). To do so, we select bandwidth bounds $\sigma_{\min}$ and $\sigma_{\max}$ and construct B logarithmically spaced bandwidths, $\sigma_b = \sigma_{\min} (\sigma_{\max}/\sigma_{\min})^{\frac{b-1}{B-1}}$, $b = 1, \dots, B$. We then sample an equal number of frequencies

from the corresponding distributions

$$\omega_{i,b} \sim \mathcal{N}(0, \sigma_b^{-2} I_d). \quad (11)$$

to get a total of $M/2$ frequencies, which then give the m test functions $\{\phi_i\}_{i=1}^M$ using the paired real features (10). If $\mathcal{X} = \mathbb{T}^d$ over the domain $[-1, 1]^d$, we sample $\omega_{i,b}$ as in (11) and then use as features $\pi \lfloor \hat{\omega}_i / \pi \rfloor$ for $i = 1, \dots, m/2$. This forces each coordinate of each of the features to be some scalar multiple of π , ensuring the corresponding test functions are periodic on $\mathbb{T}^d$.

2.5. Empirical loss

Let us now consider the empirical setting, which means we have training data $\{x_{t_k}^{(i)}\}_{i=1}^N$ in the form of samples from ρ_{t_k} for time steps $0 = t_0 < t_1 < \dots < t_K = T$. We now discuss how the loss (6) together with random Fourier feature test functions lead to an efficient train objective.

Pre-computing test-function moments and derivatives

Given m test functions $\phi_1, \dots, \phi_m$ as obtained, e.g., in the previous paragraph, we define the empirical estimator of the moment $\mathbb{E}_{\rho_{t_k}}[\phi]$ as

$$\hat{\mu}_{k,m} = \frac{1}{N} \sum_{i=1}^N \phi_m(x_{t_k}^{(i)}), \quad (12)$$

for all time steps $k = 0, \dots, K$.

There are many ways to estimate the time derivative $\frac{d}{dt} \mathbb{E}_{\rho_{t_k}}[\phi_m]$. Notice though that we use the empirical estimates (12) for estimating the time derivative, which can be noisy. We thus use a smoothing spline: For each m , fit a spline in time to $\{(t_k, \hat{\mu}_{k,m})\}_{k=1}^K$, and denote the fitted

function by $\hat{\mu}_m(t)$. The time derivative $\frac{d}{dt}\mathbb{E}_{\rho_{t_k}}[\phi_m]$ is then obtained from the spline and denoted as $\dot{\hat{\mu}}_m(t_k)$.

The moment estimate $\hat{\mu}_m$ and correspondingly their time derivative $\dot{\hat{\mu}}_m$ do not depend on the velocity field. Thus they can be pre-compute across the entire training dataset. This allows us to use all samples for estimating the moments, rather than mini-batching, which helps to reduce the variance during training iterations.

Empirical loss We now parametrize the velocity u as $u_\theta : \mathbb{R}^d \times [0, T] \rightarrow \mathbb{R}^d$ with a neural network. Let us consider the empirical loss for time steps $k = 0, \dots, K$ as

$$L_k(\theta) = \frac{1}{M} \sum_{m=1}^M \ell(\dot{\hat{\mu}}_m(t_k), \hat{\mathbb{E}}_{\rho_{t_k}}[\nabla \phi_m^\top u_\theta(t_k)]) + \lambda \mathcal{G}(u_\theta).$$

The function $\ell(x, y)$ denotes a suitable discrepancy measure. We propose to use a dual-relative loss $\ell(x, y) = \|x - y\|^2 / (\|x\|^2 + \|y\|^2 + e)$ with a small tolerance $e > 0$, which is scale-normalized and remains well-behaved when the magnitudes of different moments and their time derivatives vary widely. Such scale disparities arise naturally, for example, when a test function is not aligned well with the data and so yielding near-zero moments or derivatives or when $\|\omega_i\|$ is large, yielding disproportionately large moments or derivatives.

We then can combine the loss functions L_k over time into the an empirical loss and add a gauge regularizer to obtain our training problem

$$\min_{\theta} \frac{1}{K} \sum_{k=0}^K L_k(\theta) + \lambda \mathcal{G}(u_\theta).$$

Computation costs of random Fourier test spaces A key advantage of using random Fourier feature test function is that all required differential operators for evaluating the loss are available in closed form and are inexpensive to evaluate. In particular, every evaluation of $\phi_i(x)$, $\nabla \phi_i(x)$, or $\Delta \phi_i(x)$ requires only the inner product $\omega_i^\top x$ and simple scalar nonlinearities; the cost therefore scales linearly in the ambient dimension d ; see Appendix E for details.

3. Numerical experiments

3.1. Gigli's example

Let us revisit the example shown in Figure 1. It exhibits a family of time marginals of Gaussians with means uniformly on the unit circle. Over time, the Gaussian means rotate along the unit circle, which is an evolution that can be described by a rotational velocity field. However, if one insists on representing the time-marginal flow velocity with a gradient field, then any such gradient representation must

feature increasingly sharp gradients as the number of Gaussians grows, even though the underlying dynamics are simple; see Appendix A for details. Figure 1b shows learning the potential s_t of a gradient field $u_t = \nabla s_t$ which is in agreement with the highly oscillatory potential s_t that can explain this rotation (Figure 1a) fails. In contrast, if we learn a non-gradient velocity field with our approach (Figure 1d) using the CURL regularization, then we achieve a velocity field that is in close agreement with an analytic velocity field that explains rotation (Figure 1c). See Appendix A for details about the learning setup for this example.

3.2. Tracer particles in a turbulent field

We consider a time-dependent velocity field $v_t^{(\text{jax-cfd})}$ generated by Kolmogorov forcing as explained in [Dresdner et al. \(2022\)](#); see also Appendix D.1. We generate a training data set $\{x_{t_k}^{(i)}\}_{i=1}^N$ by integrating the ODE $\frac{d}{dt}x_t = v_t^{(\text{jax-cfd})}(x_t)$ and collect samples at the discrete times $0 = t_0 < t_1 < \dots < t_K = T$ with $x_{t_0}^{(i)}$ standard Gaussian. The generated samples correspond to tracer particles following this turbulent flow field. In accordance with the assumptions throughout this work, we do not use the trajectory information, instead selecting samples at random across different time marginals.

We derive velocity fields from the data with our approach and gauge regularizers on kinetic energy (7), divergence (9), and curl (8). Additionally, we impose gradient-structure directly by parameterizing the potential s_t and using ∇s_t in the loss (5) instead of parameterizing u_t . Figure 2(top) shows new sample populations generated with the learned velocity fields. Figure 3 shows a measure of the distance between the predicated distribution of particles and the data distribution over time. First, imposing gradient-structure by learning the potential leads to poor results. Second, notice with the freedom to use more general gauge regularizers, we can learn different particles dynamics. In particular, the field learned with the divergence regularizer (9) generates particles that behave similar to the training data particles. This is in agreement with the fact that a low divergence velocity field is a natural structural prior in this case as we know the generating vector field is divergence-free. Notice that the results obtain with the kinetic and curl regularizer look similar, which is not a coincide but the corresponding flows agree under certain conditions; see Appendix B.

3.3. Vlasov-Poisson instabilities

Let us now consider Vlasov-Poisson systems that evolve interacting charged particles; we follow the setup described in [\(Berman et al., 2024, Section B.2\)](#). In particular, we consider the two-stream instability and the bump-on-tail insta-

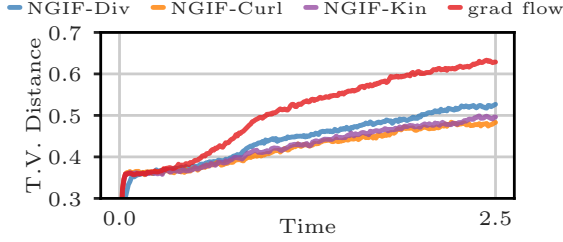


Figure 3. Total Variation distance of particle histograms over time against ground truth samples. All variants of NGIF outperform the gradient constrained version.

Table 1. Relative error in electric energy for the Vlasov-Poisson test cases. Best results are **bold**, second best are *italic*. Top 5 rows are taken from (Blickhan et al., 2025).

	two-stream ↓	bump-on-tail ↓	strong Landau ↓
CFM	1.44	5.52	1.96
NCSM	0.245	0.626	3.58
AM	0.275	0.892	NaN
HOAM	0.078	0.427	0.784
DICE	0.070	0.283	0.735
NGIF-Kin (ours)	<i>0.062</i>	0.143	0.542
NGIF-Curl (ours)	0.043	<i>0.152</i>	0.864
NGIF-Div (ours)	0.065	0.295	<i>0.701</i>

bility in two-dimensional phase space and strong Landau damping in six-dimensional phase space. The problems are parametrized by a characteristic (Debye) length parameter μ , over which we generalize as well.

The fundamental feature of these problems is a transfer of kinetic energy into potential (electric) energy, with the latter growing by several orders of magnitude throughout the simulation. The metric of interest we compute is hence the evolution of electric energy and we report the relative error in Table 1. Notice the three variants of NGIF all outperform the competing methods, most notable outperforming AM, HOAM and DICE all of which parameterize with the gradient of scalar potential. In Figure 4 we see again how different regularizers correspond to different learned sample dynamics. The divergence regularizer is able to learn particles trajectories that exhibit richer movement, which allows it to more easily develop in the final distribution.

3.4. High-Dimensional turbulence simulation

We consider the forced turbulent flow setup from Section 3.2 on the flat torus $[0, 2\pi)^2$, integrating to final time $T = 6.25$. Rather than observing advected particles, we observe cross-sectional snapshots of the vorticity field, a $d = 128 \times 128$ dimensional state. We embed these fields into a learned latent space via an autoencoder (Appendix G.5) and apply NGIF-Kin to the resulting latent marginals.

Figure 5 visualizes a rollout generated by integrating the

learned velocity field starting from an initial latent sample. This rollout should not be interpreted as recovering the true sample trajectory of the turbulent system; in this regime, individual trajectories are dominated by fast, chaotic advection and are not the target of our method. Instead, the rollout is representative in the sense that its law is designed to match the observed time-marginal evolution. To assess this population-level agreement, we track the enstrophy of each frame over time and compare its evolution to that of the true dynamics. The enstrophy curves show that NGIF accurately captures the distributional decay, demonstrating that the learned flow reproduces the salient marginal statistics even though the physical sample paths are not recovered.

This experiment highlights the distinction between sample dynamics and population dynamics in turbulence. The simulation exhibits rapid advection on short time scales alongside a slower evolution of coarse quantities such as enstrophy; population dynamics inference targets the latter.

4. Related methods

Population dynamics inference has been studied extensively in computational biology. Early approaches combined neural ODE parameterizations with snapshot matching (Hashimoto et al., 2016) or learned couplings between consecutive marginals via optimal transport (Schiebinger et al., 2019). Closer to our approach are methods that infer an admissible velocity field for a predetermined path $t \mapsto \rho(t)$; see 1 (Fixing the gauge with minimal kinetic energy).

Schrödinger Bridges The stochastic analogue to JKO methods is known as Schrödinger bridge matching (Chen et al., 2019; 2023; Shen et al., 2025; Hong et al., 2025): successive time marginals are not connected by optimal transport maps but by stochastic processes with endpoints $\rho(t_j)$ and $\rho(t_{j+1})$. The inference step turns into a stochastic differential equation (SDE) integration with the noise level taking the role of a hyperparameter. These vector fields learned in these methods obey a slightly different selection criterion (Léonard, 2014, Definition 2.1). They are also of gradient form. The variant of our method built on the Fokker-Planck equation discussed in Section F.1 is reminiscent of these.

Relation to moment matching: More broadly, our approach related to moment matching; however, instead of matching two distributions (e.g., reference and a target distribution) by regressing on test-function moments (as in kernel mean embeddings and maximum mean discrepancy (MMD, Gretton et al. (2012))) and their random-feature approximations), we match how these generalized moments change over time in a way that is consistent with transport under a velocity field. Thus, our objective is not a static

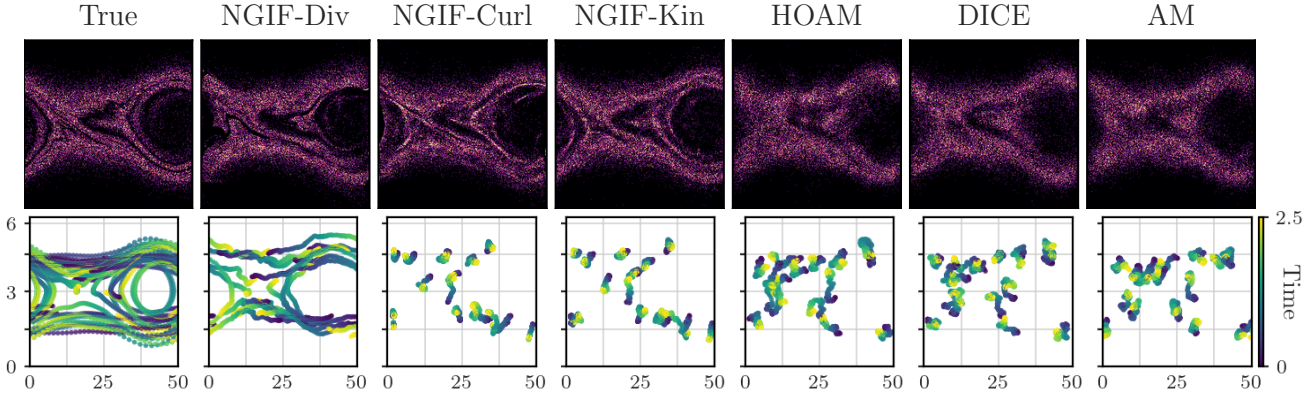


Figure 4. Vlasov-Poisson two-stream test case: Our method with any of the regularizers (7)–(9) leads to sharper, more accurate solutions fields than those which parameterize with gradient fields (HOAM, DICE, AM). Notice all variants NGIF-Curl NGIF-Kin, HOAM, DICE, AM lead to fields with low kinetic energy with little rotation as expected, while our NGIF-Div does not penalize rotational dynamics.

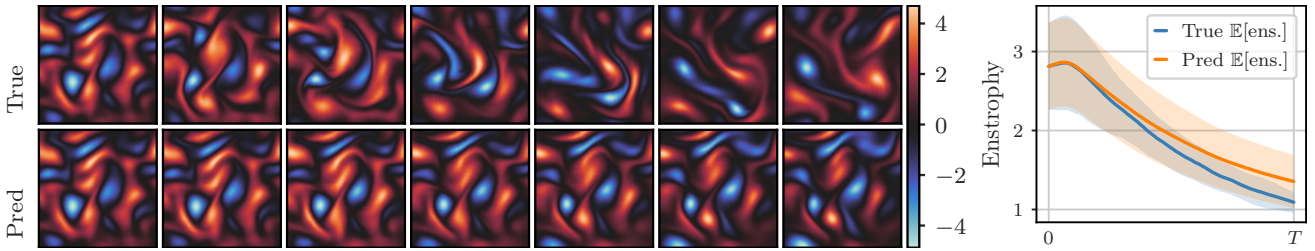


Figure 5. High-dimensional turbulence simulation: we plot a true vorticity trajectory to a rollout generated by integrating the NGIF-Kin velocity field from an initial sample (left). The NGIF rollout is not expected to match the true vorticity trajectory, since the objective is population dynamics inference from time-marginal snapshots rather than trajectory recovery. To evaluate population-level agreement, we compare the enstrophy (right), yielding a 1D distribution at each time point; its mean and standard deviation are plotted, and our NGIF accurately captures the distributional changes in enstrophy.

discrepancy aiming to connect a reference and a target distribution. Rather, it is a time-dependent constrained coming from the weak continuity equation. This perspective is complementary to recent moment-matching generative modeling (Zhou et al., 2025), which targets the static setting of sampling from a target, whereas we use moment constraints to infer population dynamics.

5. Conclusions and limitations

Conclusions. Population dynamics inference is inherently underdetermined because enforcing the continuity equation to match time marginals constrains only the flux divergence and therefore admits a whole family of velocity fields. Our main contribution is a formulation that separates marginal matching, enforced through weak (moment-based) continuity constraints, from selecting velocity field with a specific structural, inductive bias, implemented via an explicit regularizer. This decoupling makes the inductive bias a transparent modeling choice and avoids hard-wiring the minimal-kinetic-energy/gradient-field gauge, which can be inappropriate when rotational components are essential and can lead to irregular, hard-to-learn targets. Overall, the approach provides a practical route to learn population-level

dynamical models from samples of time marginals alone, supporting data-driven and reduced modeling in settings where trajectory data are unavailable or uninformative.

Limitations. (i) In practice we can enforce the weak continuity equation only against a finite collection of test functions, and it is generally unclear how large this collection must be to yield an informative constraint set for a given problem. While random Fourier tests are inexpensive to evaluate and perform well in our experiments, their informativeness can degrade in very high dimensions, potentially requiring additional prior structure. (ii) Several useful gauge regularizers depend on spatial derivatives of the learned field. Although the weak continuity equation loss avoids differentiating the network, these gauges reintroduce Jacobian computations that can become challenging in very high-dimensions. (iii) Since we optimize both the residual and the gauge criterion at the same time with relative weight λ (c.f. Equation 6), the optimization process is able to find a trade-off between these two objectives. Arguably more principled would be the formulation as a constrained optimization problem: Find $\min_u \mathcal{G}(u)$ subject to the constraint (4). This presents an interesting avenue to explore in future work.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Gigli’s example

An example of pathological gradient fields (Gigli, 2011, Example 2.4) Consider

$$\rho(x, t) = \sum_{n=1}^N \delta(x - \exp(i(2\pi n/N + \omega t)))$$

with $x \in \mathbb{R}^2$. This curve of densities describes N points equispaced along the unit circle that rotate with angular velocity ω over time. The points are located, in radial coordinates, at $r = 1$, $\theta_n(t) = 2\pi n/N + \omega t$, where $r = |x|$ and $\theta = \arctan2(x_2, x_1)$.

An admissible velocity field is given by $v(x) = \omega \Omega x$, with $\Omega = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$: It holds $\frac{d}{dt}x_t = v(x_t)$ for $x_t \sim \rho(t)$.

There also exists another admissible scalar field φ_N such that $\frac{d}{dt}x_t = \nabla \varphi_N(x_t)$, it is given by (for example) $\varphi_N(t, x) = \omega N^{-1} r^N \sin(N(\theta - \omega t))$. Indeed, we can verify $\partial_r \varphi_N(x)|_{x=x_t} = 0$ and $\frac{1}{r} \partial_\theta \varphi_N(x)|_{x=x_t} = \omega$, i.e. $\nabla \varphi_N(x)|_{x=x_t} = \omega \hat{\theta} = \omega \Omega x_t$.

This field is highly oscillatory, more so as $N \rightarrow \infty$. In this limit, we should obtain $\varphi_\infty = \text{constant}$, but it is evident that φ_N is not a sensible target of inference compared to the much more regular field v .

It is shown in (Gigli, 2011, Example 2.3) that *every* gradient vector field that is compatible with the curve $t \mapsto \rho(t)$ exhibits this behavior.

B. Relation between curl and kin regularizer

We know from OT theory that an admissible vector field is kinetic energy-optimal if and only if it is of gradient form (generalizing the concept of gradient in the case of regularity issues, see Ambrosio et al. (2005), Proposition 8.4.5). Furthermore, $\nabla u = (\nabla u)^T$ for every gradient field and the reverse implication is true on $\mathbb{R}^d$ by the Poincaré lemma (Lee, 2012, Corollary 17.15). Given the complexity of the non-linear optimization, we do not have a numerical guarantee that the two choices lead to the same vector field, but we see in the numerical experiments that they lead to very similar sample trajectories (c.f. Figures 2 and 4). The advantage of the kin regularizer is that it does not require derivative information of u and hence avoids backpropagation during optimization.

C. Formal argument why gradient fields minimize the kinetic energy

We give a formal argument for why the minimal kinetic energy field takes the form of a gradient for the convenience of the reader. A similar line of reasoning can be found in (Neklyudov et al., 2023). Consider the objective

$$\min_u \max_s \left(\int \frac{1}{2} |u_t|^2 d\rho_t(x) dt + \int (s_t \partial_t \rho_t - \nabla s_t \cdot u_t \rho_t) dx dt \right).$$

The function s plays the role of a Lagrange multiplier to enforce the continuity equation constraint. Assuming we can swap the order of max and min, we can explicitly solve for the minimal $u^{\min}$:

$$0 = \int (u_t^{\min}(x) - \nabla s_t(x)) d\rho_t(x) dt,$$

i.e. $u_t^{\min} = \nabla s_t$ ρ_t -almost everywhere. Substituting this expression back into the original objective yields

$$\max_s \int \left(\frac{1}{2} |\nabla s_t|^2 \rho_t + s_t \partial_t \rho_t - \nabla s_t \cdot \nabla s_t \rho_t \right) dx dt = - \min_s \int \left(\frac{1}{2} |\nabla s_t|^2 \rho_t - s_t \partial_t \rho_t \right) dx dt.$$

D. Details on numerical examples

D.1. Tracer particle example

The tracer particles are advected with the flow field from the `spectral_forced_turbulence` example in the `jax-cfd` project with the default choice of parameters therein. The initial ρ_0 particle distribution follows a standard Gaussian centered in the $[0, 2\pi)^2$ square. Particles that exit this box enter again following periodic boundary conditions.

D.2. Vlasov-Poisson problems

We follow the same problem set ups given in (Berman et al., 2024). In particular, the problem is parametrized by a characteristic (Debye) length parameter μ . In the considered problem, ten values of μ are considered.

The governing equation for a particle located at $x_t = [x_t^1, x_t^2]^\top$ is

$$\frac{d}{dt} \begin{bmatrix} x_t^1 \\ x_t^2 \end{bmatrix} = \begin{bmatrix} x_t^2 \\ \nabla \phi_t(x^1) \end{bmatrix},$$

where ϕ solves the Poisson equation

$$-\mu^2 \Delta \phi_t = 1 - \int \rho_t(x) dx^2.$$

The parameter μ varies as $\mu_{\text{train}} \in \{1.2, 1.3, \dots, 1.9\}$ and $\mu_{\text{test}} \in \{1.25, 1.85\}$. Computed metrics are given for the average over the test set.

The metric we consider as a quality of fit is an integrated error in electric energy given by $e = \frac{\mu^2}{2} \int |\nabla \phi_t|^2 dx^1$. We report the relative error e.e. $:= \frac{1}{T} \int_0^T |e_{\text{true}}(t) - e_{\text{predict}}(t)| / |e_{\text{true}}(t)|$. Note that the electric energy varies over two orders of magnitude throughout the simulation.

The electric potential ϕ_t is computed during integration of the dynamics by binning the samples in x^1 and solving Poisson's equation using a centered finite difference stencil.

E. Derivatives of test functions

For the standard random feature Fourier test functions with pair

$$\phi_{2i-1}(x) = \sin(\omega_i^\top x), \quad \phi_{2i}(x) = \cos(\omega_i^\top x),$$

their gradients are

$$\nabla \phi_{2i-1}(x) = \cos(\omega_i^\top x) \omega_i, \quad \nabla \phi_{2i}(x) = -\sin(\omega_i^\top x) \omega_i,$$

and their Laplacians are

$$\Delta \phi_{2i-1}(x) = -\|\omega_i\|^2 \sin(\omega_i^\top x), \quad \Delta \phi_{2i}(x) = -\|\omega_i\|^2 \cos(\omega_i^\top x).$$

F. Algorithm

F.1. Variant of our method building on Fokker-Planck

Instead of building on the continuity equation (4), we can also build on the Fokker-Planck equation,

$$\partial_t \rho_t + \nabla \cdot (\rho_t u_t^\varepsilon) = \frac{\varepsilon^2}{2} \Delta \rho_t.$$

In this case, the residual corresponding to the weak form is

$$R_t(\phi; v) = \frac{d}{dt} \mathbb{E}_{\rho_t}[\phi] - \mathbb{E}_{\rho_t}[\nabla \phi \cdot v_t(x)] - \frac{\varepsilon^2}{2} \mathbb{E}_{\rho_t}[\Delta \phi],$$

under the assumption of $\nabla \rho \rightarrow 0$ as $|x| \rightarrow \infty$ for the case $x \in \mathbb{R}^d$. The parameter ε is regularization parameter. Correspondingly, instead of the flow ODE (2), one then integrates the SDE

$$dx_t = u_t^\varepsilon(x_t) dt + \varepsilon dw_t, \quad x_t \sim \rho_0,$$

where w_t denotes a standard Wiener process, to generate new samples. In practice we find this meaningfully helps generating accurate samples. Usually ε does not need to be tuned and can just be set to $\approx 1e-2$

Algorithm 1 NGIF: Weak-form velocity inference with explicit gauge fixing

INPUTS Snapshots $\{x_k^{(i)}\}_{i=1}^N \sim \omega_{t_k}$ for $k = 0, \dots, K$; times $0 = t_0 < \dots < t_K$; bandwidths $\{\sigma_b\}_{b=1}^B$; velocity NN $u_\theta(x, t)$; gauge $G(\cdot)$; gauge weight $\gamma > 0$.

SAMPLE TEST FUNCTIONS

Choose an even number of tests m and sample frequencies $\{\rho_j\}_{j=1}^{m/2}$ using the given bandwidths $\{\sigma_b\}$.

Define $\vartheta_{2j-1}(x) = \sin(\rho_j^\top x)$, $\vartheta_{2j}(x) = \cos(\rho_j^\top x)$, with $\nabla \vartheta_{2j-1}(x) = \cos(\rho_j^\top x) \rho_j$ and $\nabla \vartheta_{2j}(x) = -\sin(\rho_j^\top x) \rho_j$.

PRECOMPUTE MOMENTS AND TIME-DERIVATIVES

For all k, r : $\hat{\mu}_{k,r} \leftarrow \frac{1}{N} \sum_{i=1}^N \vartheta_r(x_k^{(i)})$.

For each r , fit a smoothing spline $\hat{\mu}_r(t)$ through $\{(t_k, \hat{\mu}_{k,r})\}_{k=0}^K$ and set $\tilde{\mu}_{k,r} \leftarrow \frac{d}{dt} \hat{\mu}_r(t)|_{t=t_k}$ for all k .

TRAINING

while not converged **do**

 Sample a single time index $k \sim \text{Unif}\{0, \dots, K\}$ and a minibatch $\mathcal{B} \subset \{1, \dots, N\}$

 For all r : $\hat{T}_{k,r}(\theta) \leftarrow \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \nabla \vartheta_r(x_k^{(i)})^\top u_\theta(x_k^{(i)}, t_k)$

$\ell(a, b) \leftarrow \frac{\|a-b\|^2}{\|a\|^2 + \|b\|^2}$

$\mathcal{L}_k(\theta) \leftarrow \frac{1}{m} \sum_{r=1}^m \ell(\tilde{\mu}_{k,r}, \hat{T}_{k,r}(\theta)) + \gamma G(u_\theta)$

 Update $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_k(\theta)$ {e.g. Adam}

end while

Output. Learned velocity field $u_\theta(x, t)$.

G. Implementation details

G.1. Data preparation and normalization

All input variables are normalized independently to lie in $[-1, 1]$ along each dimension using statistics computed from the training set. Model training is performed in this normalized space. For evaluation, we apply the inverse normalization to model outputs and compute all reported metrics in the original data domain. This normalization is important in practice, as it allows the same kernel bandwidths (e.g., for RFF features) to be applied consistently across dimensions and datasets.

G.2. Computing $\dot{\mu}$ via Splines

We estimate smooth moment trajectories and their time derivatives by fitting a smoothing spline to each empirical moment time series. Given observation times t_{data} and moment values $\mu(t_{\text{data}})$, we construct $s(t) = \text{make_smoothing_spline}(t_{\text{data}}, \mu; \nu)$ (SciPy) and evaluate $\hat{\mu}(t_{\text{data}}) = \frac{d}{dt} s(t)|_{t=t_{\text{data}}}$. Unless otherwise stated, the spline regularization parameter is fixed to $\nu = 10^{-5}$ in all experiments. As noted in the main text, these spline fits and derivative evaluations are precomputed once on the full dataset prior to training.

G.3. RFF bandwidth selection

Performance is sensitive to the choice of kernel bandwidths (RFF σ values). We therefore use a multiple-bandwidth approach, which consistently performs better than a single scale. A reasonable starting point for σ is from applying the median heuristic to the training data to obtain a base bandwidth σ_{med} , and include additional scales one order of magnitude above and below, i.e., $\{\sigma_{\text{med}}/10, \sigma_{\text{med}}, 10\sigma_{\text{med}}\}$.

G.4. Hyperparameters

For all experiments unless otherwise noted the following algorithmic hyperparameters were used,

	Gigli	Tracer particles	Vlasov-Poisson	High-Dim Turbulence
$b_{\min}$	0.05	0.05	0.05	4.0
$b_{\max}$	0.05	1.0	1.0	16.0
B	1	100	100	100
ν	10^{-5}	10^{-5}	10^{-5}	10^{-5}
ε	0.0	10^{-2}	10^{-2}	0.0
M	50,000	50,000	50,000	50,000

	Tracer particles	VP (two-stream, bump)	VP (strong-Landau)	High-Dim Turbulence
λ_{kin}	1e-2	1e-2	1e-2	1e-2
λ_{curl}	1e-2	1e-3	5e-3	—
λ_{div}	1e-2	1e-3	5e-3	—

Hyperparameter	Value
Learning rate (lr)	5×10^{-4}
Iterations (iters)	500,000
Scheduler	cos
Optimizer	Adam
Precision	float32

G.5. Neural Net Architectures

Autoencoder Architecture. Convolutional encoder/decoder with Swish + GroupNorm (8 groups) and dropout 0.05. Four downsampling stages $128 \rightarrow 64 \rightarrow 32 \rightarrow 16$ (strided conv) mirrored by four upsampling stages (transpose conv). Base channels 32, doubling each downsample (bottleneck 512). Spatial latent with posterior $(\mu, \log \sigma^2) \in \mathbb{R}^{16 \times 16 \times 1}$; $\log \sigma^2$ clipped to $[-12, 6]$.

Loss (KL-realized). Reconstruction = MSE + 0.10 MAE plus β -KL regularization with $\beta = 5 \times 10^{-4}$. Linear KL warmup over 5000 optimizer steps.

Latent post-processing. After training, per-dimension whitening stats of μ computed over the dataset; encoding uses μ by default and returns whitened latents optionally tanh-squashed to $[-1, 1]$ (enabled; scale 2.0). Decoding inverts squash (atanh) and whitening.

For all other decisions we use standard, best-practice choices for convolutional VAE training and evaluation.

MLP Backbone All parameterizations of u_θ except the high-dimensional ones are done using a MLP.

For periodic problems, we embed x using harmonic features with $H = 4$:

$$\phi(x) = [\sin(\omega_0 m x), \cos(\omega_0 m x)]_{m=1}^4, \quad \omega_0 = \frac{2\pi}{\text{period}},$$

and concatenate features across dimensions. For non-periodic problems, the raw input is used.

The network optionally conditions on time t and auxiliary variables μ . Each is embedded by a two-layer MLP with hidden width equal to the main network width and GELU activations, producing vectors in $\mathbb{R}^{d_{\text{out}}}$. If both are present, the embeddings are concatenated. If only one is present, it is used directly.

The backbone consists of L fully connected layers of width W with GELU activations. At each layer, a conditioning bias is added. The bias is produced by passing the conditioning vector through a separate two-layer MLP of width W and adding the result to the pre-activation output.

Unless otherwise stated, we use: $W = 196$, $L = 7$, GELU activations, bias terms in all layers, period set per task, and task-dependent output dimension.

UNet backbone For the high-dimensional experiments we employ a Unet. We follow the encoder-decoder UNet with skip-connections. All architecture decisions are standard. We use GroupNorm, no dropout, stride-2 3×3 convolution. At each spatial scale for our residual blocks the channel depths are, $[128, 128, 360]$

H. Metrics

H.1. Total variation distance between binned particle measures.

Let $\{\mathcal{B}_k\}_{k=1}^B$ be a partition of $\mathbb{T}^2 = [0, 2\pi)^2$ into $B = B_x B_y$ equal bins. Given particles $\{x_n\}_{n=1}^N$, define counts and normalized histogram

$$c_k = \sum_{n=1}^N \mathbf{1}\{x_n \in \mathcal{B}_k\}, \quad p_k = \frac{c_k}{\sum_{\ell=1}^B c_\ell}.$$

For two particle sets with histograms $p, q \in \mathbb{R}^B$, the total variation distance is

$$d_{\text{TV}}(p, q) = \frac{1}{2} \sum_{k=1}^B |p_k - q_k| = \frac{1}{2} \|p - q\|_1.$$